FIELD MANUAL

OPERATING NOTES // XX29 / FM-1

XENOLITH

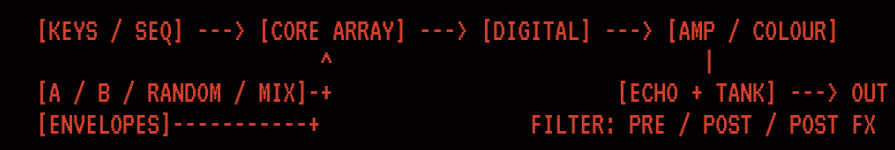
One played note. A changing array of coupled oscillators.

Xenolith begins with the eight drifting FM pairs of the SuperCollider patch. The instrument can grow to 64 cells, connect their carriers, change the calculation itself, and send the result through one shared amplifier and delay tank. The four HOME controls remain the entrance.

01 / FIRST SOUND

Press a key. Release it to hear the envelope release. With SEQ disabled, PLAY latches a drone; PLAY again stops it. Begin quietly, then set the master volume. HOME leaves a page without restoring defaults.

The starting sound is 84 Hz, index 2.4, ratio 1.2, drift 0.20 and eight cells. The core runs at 44.1 kHz; the tank is 25% wet. Modulation, sequencing, filter and amplifier colour begin off.



02 / PANEL MAP

HOME	Tonic, index, ratio, drift. Hold for array structure.
ALGORITHM	Cell count. The small HOME drawing follows the connections.
ARP	Assign keys, choose their range, or select cell bands.
SEQ	Notes, four control lanes, recording and note-gate behavior.
LFO	Rise/fall functions, random source and modulation mixer.
ENV	Four ADSRs. Hold for the selected envelope's destination.
EDIT	Digital calculation / filter. Hold for filter circuit.
FX	Tank controls. Hold for input and internal diffusion.
GLO	Spectrum. Hold for oscillator table calculation.
SELECT	On HOME: amplifier colour / shape / HOME.

03 / KEEP THE SOUND; LEAVE THE PAGE

Short HOME returns to the main panel while sound and control motion continue. REC records notes and parameter gestures, not an audio file. HOME + REC is the panic chord: it stops transport and clears the sound and tail, while keeping the take already recorded.

The knobs store a base sound. Keys, sequence lanes and modulation can move the effective values you see on HOME. A held key wins over the matching sequencer assignment; modulation offsets are added afterwards.

[READ IN ORDER, OR OPEN THE RELEVANT PLATE]

02 ARRAY / 03 KEYS + SEQ / 04 MOD / 05 FILTER + ENV
06 EXHAUST / 07 DIGITAL + TABLE / 08 COLOUR + PATCHES

CORE ARRAY

CELLS / CONNECTIONS / MODULATION

01 / FOUR HOME CONTROLS

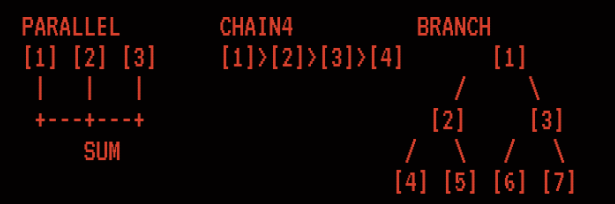
K1 TONIC 20-800 Hz. The central pitch of the array.
K2 FM INDEX 0-64. In LINEAR, deviation relative to each cell's tonic.
K3 MOD RATIO 0.001-32. Modulator frequency divided by cell tonic.
K4 DRIFT 0-1. Independent slow pitch movement in the cells.

Ratio has fine movement around familiar musical relationships, with larger steps higher up. Very low ratios turn the modulator into slow motion. Drift 0.20 is the starting wander, nominally about +/-2%. Index travel becomes coarser above 8, 16 and 32; the range stays 0-64.

ALGORITHM adds or removes cells. Each cell contains a modulator and a carrier; eight cells contain sixteen oscillators. They share the note and main envelope. More cells first thicken the cluster, then extend the frequency field. They are not separate polyphonic keyboard voices.

02 / HOLD HOME: HOW THE ARRAY GROWS

K1 GROWTH PARALLEL / CHAIN4 / BRANCH / LOOP4 / SELF
K2 MODE LINEAR / PHASE / EXP / PULSE FM / PHASE WARP
K3 LINK Strength of the additional inter-cell connection, 0-4.
K4 FEEDBACK Delayed return strength, 0-0.95, for LOOP4 or SELF.



Every cell still contributes to the sum. CHAIN4 starts a new group after four cells. BRANCH gives each parent two descendants. LOOP4 adds a delayed return from the fourth carrier to the first of a complete group. SELF feeds each carrier's previous sample back into itself. Turning ALGORITHM down removes the most recently added end of the array.

03 / FIVE DIFFERENT OPERATIONS

LINEAR changes signed instantaneous frequency, including through zero. PHASE adds directly to the carrier's lookup angle. An isolated steady sine can resemble linear FM when their phase excursions are matched; ratio changes, envelopes and feedback make that comparison less simple. EXP changes frequency multiplicatively. Its index is 0-120 semitones, beginning at one semitone; greater depth can also raise the mean pitch. PULSE FM replaces the local sine modulator with a bipolar rectangle. PHASE WARP bends the phase address, altering the carrier's shape.

[A USEFUL FIRST COMPARISON]

Hold one note, reduce drift, and compare eight PARALLEL cells with four CHAIN4 cells. Raise LINK slowly. Added connections change how spectra are generated; a larger stack mainly adds independent beating and width.

KEYS + SEQUENCE

ASSIGN / RECORD / REASSIGN

01 / ARP IS THE KEYBOARD ASSIGNMENT PAGE

K1 DESTINATION Tonic, index, ratio, clocks, bits, filter and more.
K2 KEY MODE   GLOBAL or ANS CELLS.
K3 LATCH   Keep a scalar assignment after the key is released.
K4 KEY GATE   Let key edges operate the main amplitude envelope.

Press SELECT for the range page. GLOBAL has LOW / HIGH / RESPONSE; LINEAR or LOG spreads that range across 27 keys. TONIC/CHROMATIC keeps ordinary note and octave behavior. LOG needs a positive lower value. Turning KEY GATE off is useful when keys should reshape a PLAY drone.

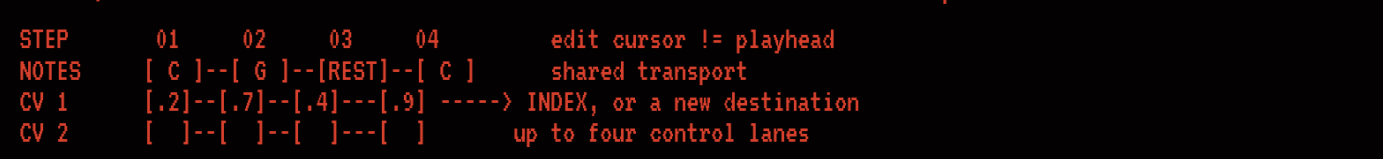
ANS CELLS selects bands within the array. Its second page sets LOW/HIGH spectral limits, BAND WIDTH and RELATIVE/SPECTRAL tuning. Held bands are combined. All 27 keys cover the array; SPECTRAL spreads cells between the chosen frequencies. The master envelope remains shared.

02 / SEQ: ONE NOTE TRACK AND FOUR CONTROL LANES

Rotate SELECT: SYSTEM / TONIC NOTES / CV 1 / CV 2 / CV 3 / CV 4.
Press SELECT: alternate the selected object's two pages.

SYSTEM: enable, tempo, length 1-16, clock division. Its second page chooses recording mode, armed material and ADSR / CONTINUOUS note gate. ADSR retriggers each NOTE and uses its gate percentage. CONTINUOUS supplies a drone while transport runs, even through rests or CV-only patterns; amplifier colour and amplitude modulation still operate.

TONIC NOTES: select step, REST / NOTE / TIE, note and gate length. The second page can trigger function A, function B and RANDOM NEXT. CV main: TARGET / ENABLE / SLEW / VALUE. CV detail: EDIT STEP / VALUE / STEP ON / FILL. Each lane owns one destination. SLEW smooths its steps.



03 / TYPE A TAKE ON THE KEYS

1. Stop transport. Set SEQ length 4. Open ARP and choose the assignment.
2. For notes use TONIC/CHROMATIC; for CV choose INDEX, RATIO or another scalar destination. Press REC there to arm the matching material.
3. Press four fresh keys. Each writes one step and advances the cursor.
4. REC ends the take. PLAY auditions it. SAVE stores the complete patch.

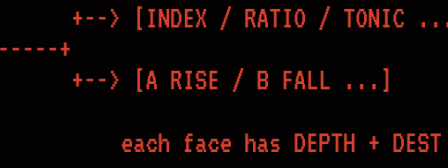
You can edit each step afterwards. A knob writes the current stopped recording step without advancing; an assigned key writes and advances. Changing a CV lane's target keeps its normalized shape. ANS CELL stores one band position per step, not a whole simultaneous keyboard mask. Running REPLACE/OVERDUB records one loop. STEP entry needs stopped play.

MODULATION

TWO FUNCTIONS / RANDOM / ONE MIXER

01 / FOUR FACES, TWO PAGES EACH

Press LFO. Rotate SELECT to A, B, RANDOM or MIX. Short LFO alternates the selected face's pages. SELECT press bypasses the entire modulation system. A real destination creates its route and enables MOD; NONE removes that face's primary route. HOME leaves the patch running.



02 / RISE AND FALL ARE THE BASIC MATERIAL

A/B main: RISE / FALL / signed DEPTH / DESTINATION.
A/B detail: MODE / INPUT / TRIGGER / CURVE.
CYCLE repeats a rise and fall. AD makes a triggered contour. GATE follows a gate; SLEW follows an input with separate rising and falling times. OFF disables that function. The same units can therefore move index, soften a sequence or turn another source into a changing contour.

RANDOM main: RATE / SPREAD / DEPTH / DESTINATION.
RANDOM detail: CORRELATE / LEVELS / CLOCK / STEP-WALK output.
Hold LFO on RANDOM: LAW / MEMORY / RENEW % / BIAS.
CURRENT keeps the original law; BELL favors central values, EDGE the ends, BINARY two endpoints. MEMORY captures 2-32 events. RENEW 0 repeats the captured phrase; 100% replaces every visited event. A small value lets the phrase change gradually. Captured events are not saved.

MIX main: INPUT A / INPUT B / DEPTH / DESTINATION.
MIX detail: GAIN A / GAIN B / OFFSET / OFF-SUM-PRODUCT-MAX.
The mixer combines control signals. It is not a second audio mixer.

03 / TWO SMALL PATCHES

MOVING INDEX: A in CYCLE, rise 0.05 s, fall 0.8 s, destination INDEX. Start with low depth. Send RANDOM WALK to FUNC A FALL, about +2 octaves, and choose FUNCTION END as its clock. Each ending chooses the next fall.

OPENING AMPLITUDE: leave A's own destination NONE. MIX input A=FUNC A, gain A=1, gain B=0, offset=-1, mode SUM; route MIX to AMPLITUDE, depth +1. PLAY with SEQ off supplies a drone. A now opens and closes its level. Direct positive A-to-amplitude at the default unity base mostly clips at full level; negative depth instead makes a useful ducking contour.

[DEPTH UNITS ARE PART OF THE PATCH]

Pitch depth is semitones; ratio and time/rate depths are octaves. Index depth 1 spans 64 index units. Bits use signed bit offsets; CELLS adds or removes cells. Core/wet clock, echo and cell changes are deliberately slow control transactions. Their base requests return after bypass.

CONTOUR + FILTER

FOUR ENVELOPES / ONE COMMON CIRCUIT

01 / ENV: ATTACK / DECAY / SUSTAIN / RELEASE

Rotate SELECT to AMP, FILTER, INDEX or TONIC. AMP is the shared amplitude envelope. The other three names are starting roles, not fixed wiring: hold ENV for DESTINATION / DEPTH / TRIGGER / ENABLE.

Choose KEY, SEQ NOTE or BOTH triggering. KEY includes the latched PLAY drone, which holds sustain until stopped. New contours begin disabled at zero depth. Start with FILTER at a modest positive octave depth, then shorten its decay. AMP's route is fixed; the other routes are free.

02 / EDIT: DIGITAL / FILTER / HOME

FILTER: K1 CUTOFF / K2 MODE / K3 NETWORK / K4 RESONANCE.
Rotate SELECT to OFF, PRE DIGITAL, POST DIGITAL or POST FX.

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CORE ---> [PRE] ---> QUANT / RECON ---> [POST] ---> AMP ---> FX ---> [POST FX]
           mono / core clock          mono / 44.1k              stereo / 44.1k
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PRE filters before the digital reduction. POST DIGITAL also filters its newly introduced roughness. POST FX shapes the complete stereo sound and tail. Choose placement first; cutoff and resonance then follow the same keys, envelopes and control lanes as other parameters.

LP / BP / HP / NOTCH / PHASE use different outputs of a common reduced triode / parallel-T feedback model. TWIN T returns its selective plate response; CUTOFF becomes CENTRE. The curvature sits inside feedback, so level changes affect the response. It is not a measured valve replica.

NETWORK selects one section, two in cascade, or a BRIDGE arrangement. The displayed names are 2POLE / 4POLE / BRIDGE, or ONE / TWO / BRIDGE in TWIN T. More sections can strengthen coloration and cancellation. For TWIN T, try centre 200 Hz and resonance 0.7 on the 84 Hz drone.

03 / HOLD EDIT: CIRCUIT

K1 CATHODE	470-10,000 ohm; default 2,200. Changes local degeneration in the loaded stage and its available feedback.
K2 HEADROOM	0.0625-1; default 0.25. Lower values bend at smaller input levels, with small-signal gain compensated.
K3 COUPLING	0.5-100 Hz; default 3.18. Raises the low-frequency coupling pole; higher settings remove more bass.
K4 T BALANCE	0.5-1; default 1. Moves linked passive component ratios; reducing balance fills the feedback network's notch.

These controls alter the shared cell beneath every response. Return to the ordinary filter page to hear their interaction with resonance and placement. The book supplies the parallel-T and feedback circuit; the selectable taps, parameter ranges and digital reduction are ours.

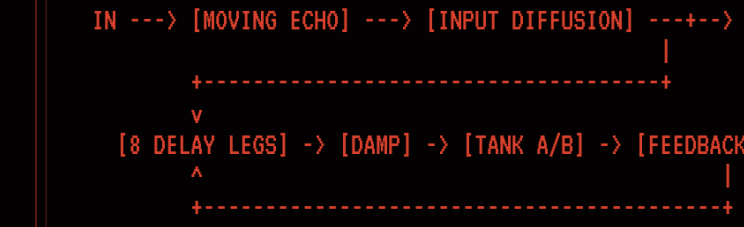
EXHAUST BUS

ECHO / FEEDBACK / DIFFUSION

01 / FX: THE FIRST TWO PAGES

K1 ECHO ms Moving delay before the tank. 2 ms to available memory.
K2 DAMP Hz High-frequency absorption in the recirculating paths.
K3 TAIL s Nominal low-frequency decay time of the feedback tank.
K4 MIX Dry/wet balance.

Press FX again: WET kHz / STRUCTURE / DIFFUSION / WIDTH.
Press SELECT on an FX page to change HOLD / LINEAR reconstruction.
Short FX from the second page returns HOME. Hold FX opens diffusion.



STRUCTURE changes how the stored sound is connected or heard. TANK is the base network. CLOUD4 / CLOUD8 expose additional early taps. RING circulates between legs; PAIRS couples them in pairs. The same delay memory continues through the change. WIDTH reduces the stereo spread toward mono; DIFFUSION softens the input before it enters the tank.

02 / HOLD FX: INSIDE THE TANK

K1 INPUT The same input-diffusion control, 0-85%.
K2 TANK A Short allpass sections in the first four feedback legs.
K3 TANK B The corresponding sections in the other four legs.
K4 SCALE Quarter, half or full short-delay lengths.

A/B begin at zero. Raise one group first and listen to the tail after stopping a note. These sections redistribute delay across frequencies; they change the recirculating modes without being ordinary distortion. Changing SCALE briefly retires their small histories. The long tail remains. SCALE is a delay geometry, not a literal room-size control.

03 / THREE DIFFERENT WAYS TO MOVE TIME

ECHO sweeps move a read head. Its changing distance bends pitch and comb resonances, much like changing the speed of a recorded passage. Those bends do not by themselves mean aliasing.
WET clock spans about 1.38-44.1 kHz. Lowering it changes sampling colour and stretches the fixed tank paths. ECHO can reach 1 s at 44.1 kHz, 2 s at 22.05 kHz, and up to 32 s at the slow end.
TAIL changes recirculation strength. It does not set the echo spacing. Start with a short note, long tail and modest mix; move only one time control until its role becomes audible. Then let a slow function move it.

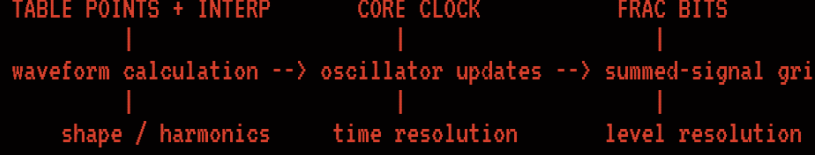
DIGITAL + TABLE

CLOCK / QUANTIZATION / WAVEFORM

01 / EDIT: THE DIGITAL PAGE

K1 MAX kHz Requested core rate: 1-44.1 kHz, then 88.2 / 176.4.
K2 FRAC BITS Native bypass at 24; turn down through the fractional grid.
K3 QUANT TRUNC / ROUND / ERROR FB.
K4 RECON HOLD (ZOH) / LINEAR for lower-rate reconstruction.

The output remains 44.1 kHz. A higher internal rate computes more FM steps before filtering back down. A lower rate creates fewer updates, then holds or interpolates them. Cell count and active processing limit the admitted rate. MAX is remembered: the faster request returns when the work that required a lower ceiling has retired.



24 is native bypass, not proof that every calculation uses a 24-bit number. Values 2-23 quantize the summed signal on a fractional grid; they do not reduce the precision of every oscillator operation. At 1, the output becomes a polarity signal: -0.5 or +0.5, with exact zero retained. ERROR FB carries a bounded quantization error into later samples, making density part of the sound. Bits are also a destination.

02 / HOLD GLO: OSCILLATOR TABLE CALCULATION

K1 TABLE POINTS 4-2048, powers of two; default 2048.
K2 INTERP HOLD / LINEAR; default LINEAR.
K3 HARMONIC 0-100%; blend toward a third-harmonic transformation.
K4 TARGET MOD / CAR / BOTH; choose which oscillator is altered.

Fewer points coarsen the waveform before it participates in FM. Four points with LINEAR form a triangle; HOLD makes a staircase. HARMONIC transforms an ideal sine toward its third harmonic; coarse tables add their own extra components. The carrier's pitch can seem to triple at 100%. Target MOD to reshape the excitation, or CAR to alter the result. PULSE FM already uses a rectangle for its modulator, so MOD table changes do not reshape that rectangle. The carrier remains available.

03 / GLO: OBSERVE THE RESULT

The spectrum and phase display measures audio. PRESETS chooses L/R/MID; GLO then EDIT opens SPAN / TAP / FFT FPS / FFT SIZE. SCL toggles the spectrum-driven note LEDs. Dense sidebands are generated by the paired oscillators; they are not individually allocated voices. More internal rate can reduce folding, but neither interpolation nor oversampling makes every high-index, feedback or warped setting free of aliasing.

COLOUR + MEMORY

FINISH / SAVE / RETURN

01 / SELECT FROM HOME: AMPLIFIER

First page: DRIVE / BIAS / COLOUR / actual AMPLITUDE, read-only. Press SELECT again: CURVE / HP Hz / TONE / GAIN dB. Again returns HOME. COLOUR blends a separate processed branch; zero bypasses that branch exactly. Its high-pass, tone filter and gain therefore act only where colour is present. Start with DRIVE, then balance with COLOUR and GAIN.

CUBIC supplies the original soft curve. SPACE3/2 uses asymmetric three-halves-law curvature. SOFTSIGN compresses progressively. FOLD reflects peaks rather than flattening them; drive adds further folds. BIAS changes the operating position on these curves. These are bounded digital transfer functions, not four different complete valve circuits. The VCA FOLD modulation destination moves DRIVE when FOLD is selected. Keep COLOUR above zero to hear it; depth is a signed normalized offset.

02 / SIXTEEN PATCH SLOTS

PRESETS chooses 01-16 without immediately loading. PLAY on the patch page loads an occupied slot. SAVE, choose the slot, SAVE confirms. HOME cancels an unconfirmed save. Recall starts transport and REC stopped.

A patch stores sound, array, key assignment, envelopes, modulation, sequence and effect settings. Held keys, oscillator phases, tank audio and the actual captured random-memory phrase are not part of the patch. Save a useful experiment before changing firmware or backing up.

Open.bat opens the PC instrument. Update.bat is the local installer. Backup.bat saves settings and all sixteen slots to a dated archive; restore validates the archive and asks for physical SELECT confirmation. The device's HOME button cancels that restore prompt.

03 / A FEW USEFUL DIRECTIONS

GLASS: a small array, high allowed rate, restrained drift, slow index and ratio movement. Listen to sidebands crossing before adding colour.
MASS: grow the parallel array, then compare BRANCH at low LINK. Leave enough space in the tank to hear beating before increasing feedback.
PULSE: reduce to one bit, then modulate index or ratio. The sign changes become a complex pulse pattern; a POST filter can reveal its lower shape.

04 / THE READING BEHIND THE CONTROLS

SMITH: sinusoidal FM and the DFT explain sidebands and their measurement. The oscillator pair generates the expansion; the FFT observes it later.
WHITE / EARLY DSP: finite-wordlength arithmetic and error feedback inform the quantizer. Precision and clock are separate choices, with audible costs.
DATTORRO + SMITH: input versus tank diffusion, and normalized allpass sections, inform the delay network. This tank is Xenolith's arrangement.
RADIOTRON + SEELY: loaded stages, degeneration and feedback provide the common filter's reduced circuit and the amplifier curve's limited lineage.

The engineering notes retain equations, sources and test scope. This manual is for the panel. Keep the base sound, change one relation, listen to the consequence, then save it. The apparatus is ready to be used.